



**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**

CS23432 – Software Construction

(REGULATION 2023)

**RAJALAKSHMI ENGINEERING COLLEGE,
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EXP NO: 1

AZURE DEVOPS ENVIRONMENT SETUP

Aim:

To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

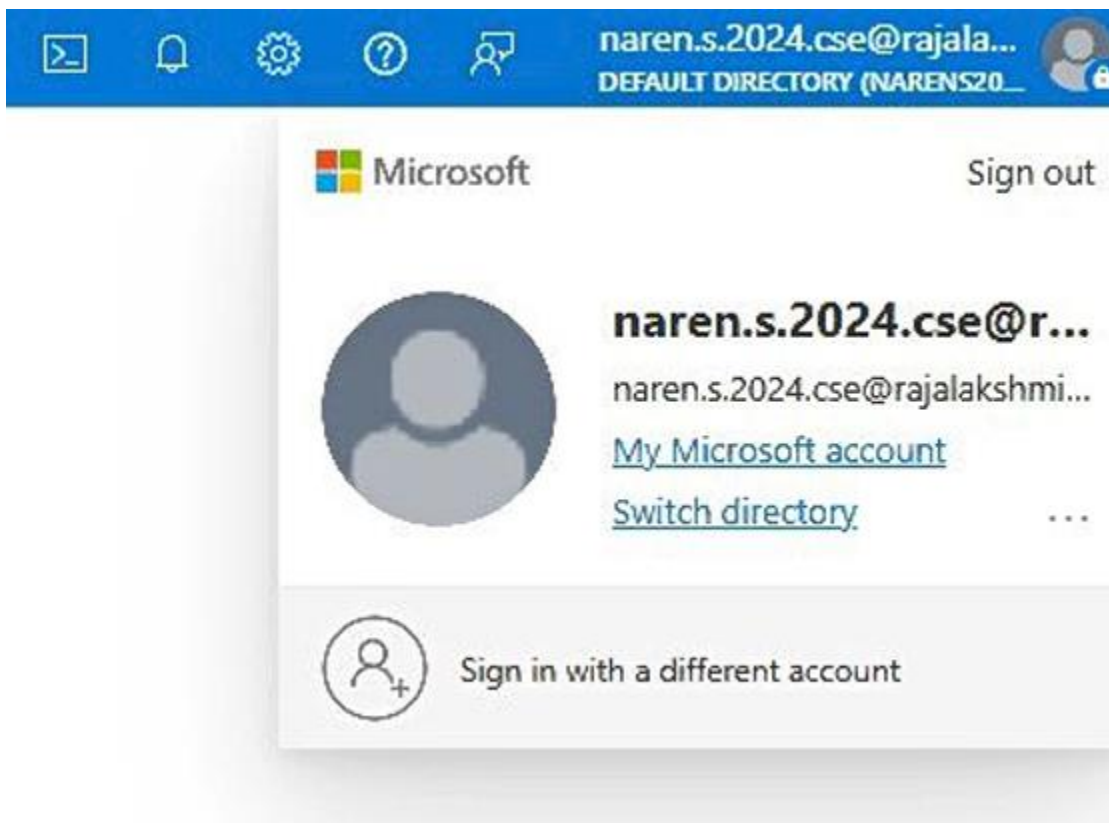
INSTALLATION

1. Open your web browser and go to the Azure website: <https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

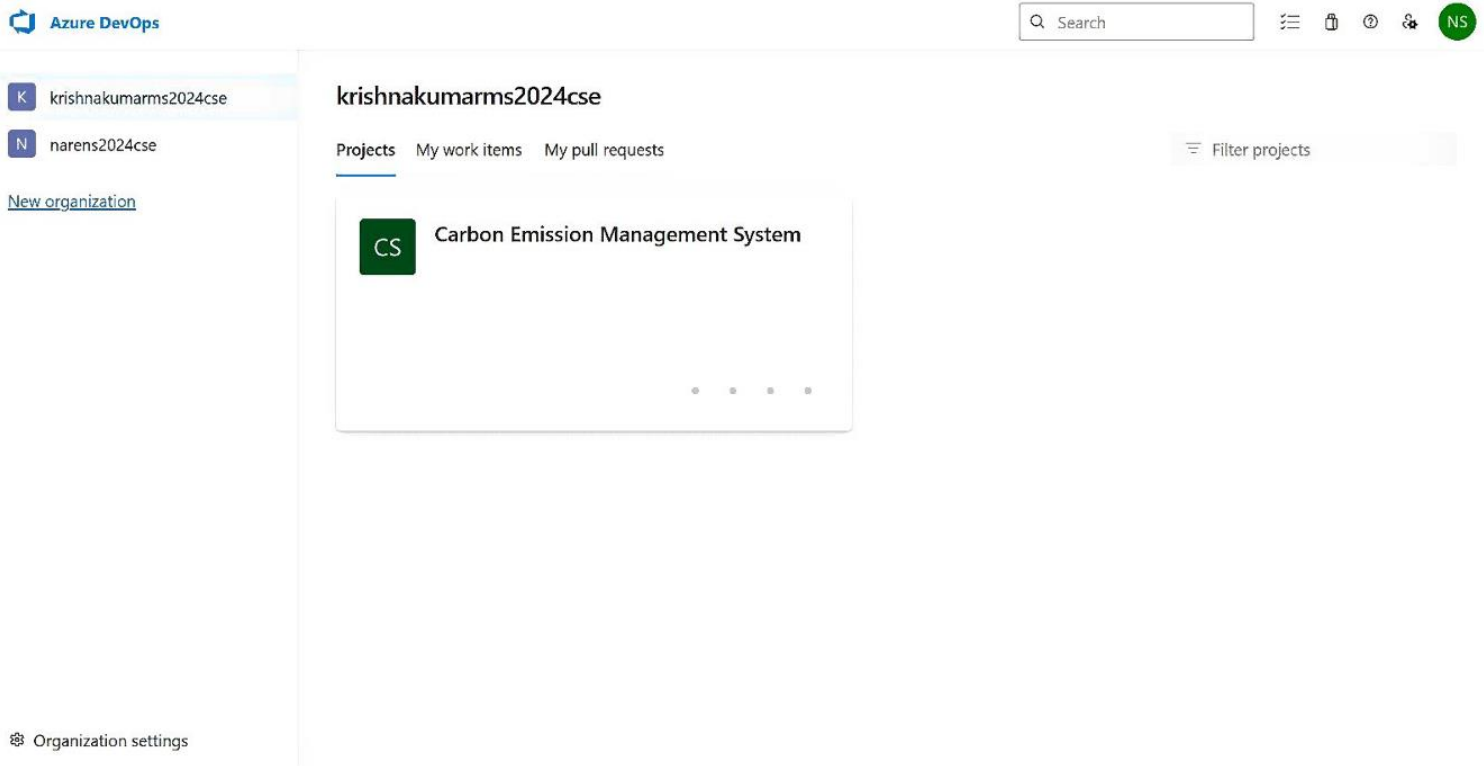
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>

2. Azure home page



3. Open DevOps environment in the Azure platform by typing *Azure DevOps Organizations* in the search bar.

4. Click on the **My Azure DevOps Organization** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2

AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

Aim:

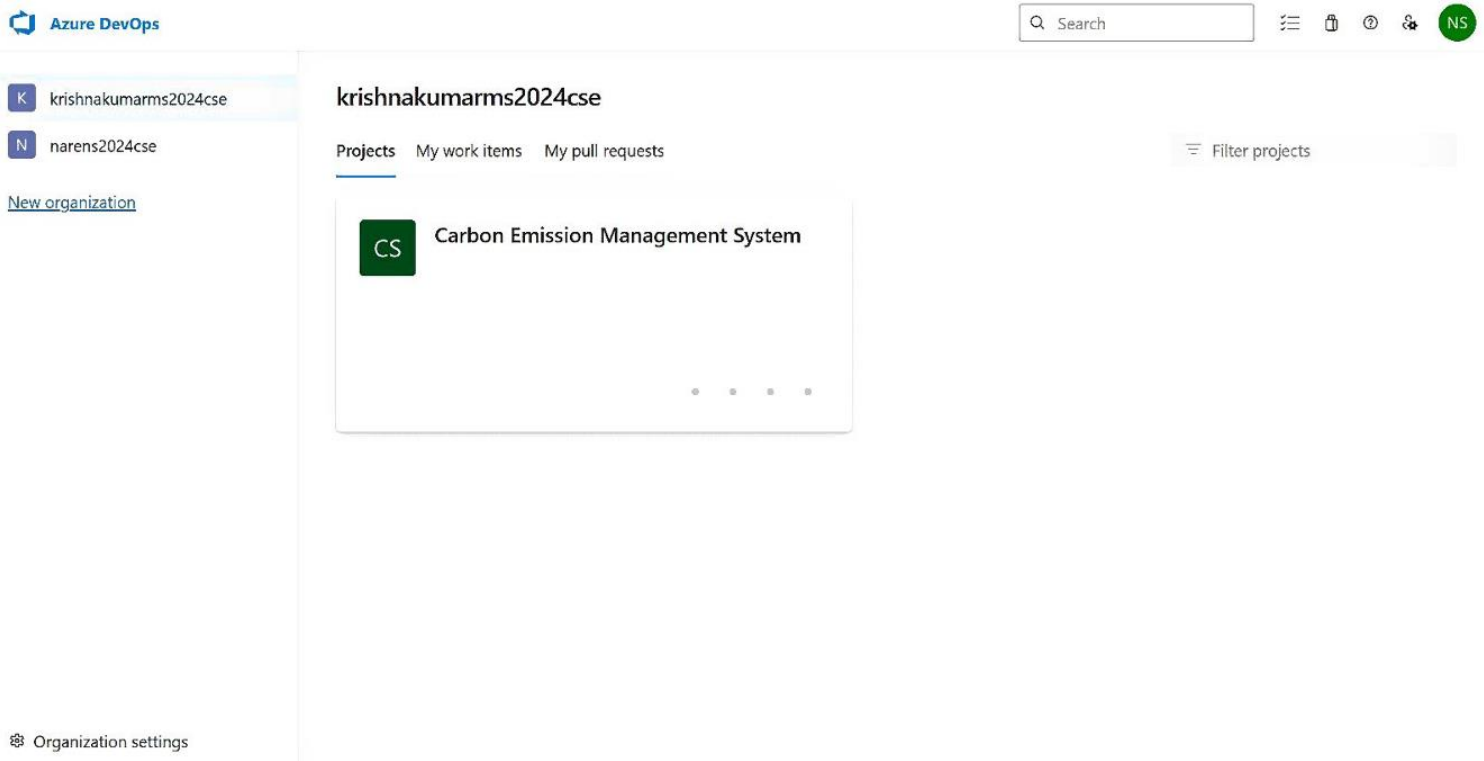
To set up an Azure DevOps project for efficient collaboration and agile work management.

1. Create An Azure Account

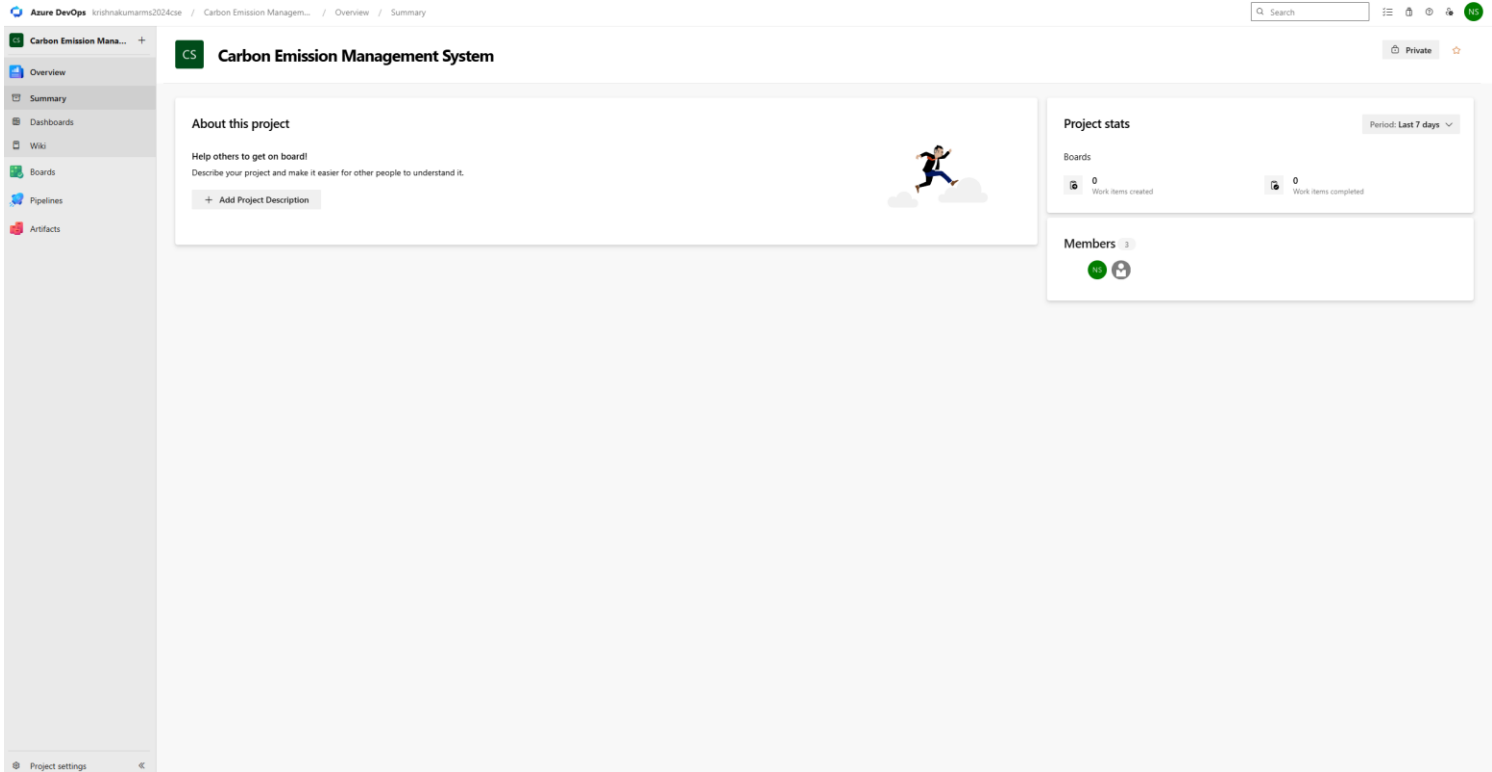
2. Create the First Project in Your Organization

- After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- On the organization's **Home page**, click on the **New Project** button.
- Enter the project name, description, and visibility options:
 - Name:** Choose a name for the project (e.g., **LMS**).
 - Description:** Optionally, add a description to provide more context about the project.
 - Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- Once you've filled out the details, click **Create** to set up your first project.

3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.



4. Project dashboard



5. To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

Azure DevOps krishnakumams2024ce / Carbon Emission Managem... / Boards / Work items

Carbon Emission Mana... +

Overview

Boards

Work Items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Pipelines

Artifacts

Project settings

Work items

Recently updated ▾ + New Work Item ▾ ↗ Open in Queries ↗ Column Options ↗ Import Work Items ↗ Recycle Bin ↗

Filter by keyword

Types ▾ Assigned to ▾ States ▾ Area ▾ Tags ▾ X

ID	Title	Assigned To	State	Area Path	Tags	Comments	Activity Date
1	User Management System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:36:01 PM
2	User Registration	Unassigned	New	Carbon Emission Management...			4/27/2026 4:35:53 PM
3	User login and authentication	Unassigned	New	Carbon Emission Management...			4/27/2026 4:35:39 PM
4	Profile Management	Unassigned	New	Carbon Emission Management...			4/27/2026 4:35:26 PM
5	As a user, i want to login securely using my credentials	Unassigned	New	Carbon Emission Management...			5/2/2026 3:27:21 AM
6	As a user, i want to update my profile information	Unassigned	New	Carbon Emission Management...			4/27/2026 4:34:55 PM
7	Carbon Emission Tracking System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:34:37 PM
8	Activity Data Input	Unassigned	New	Carbon Emission Management...			4/27/2026 4:34:27 PM
9	Emission Tracking Module	Unassigned	New	Carbon Emission Management...			4/27/2026 4:34:14 PM
10	Real-Time Data Update	Unassigned	New	Carbon Emission Management...			4/27/2026 4:33:52 PM
11	As a user, i want to enter my daily activities to track emissions	Unassigned	New	Carbon Emission Management...			4/27/2026 4:33:39 PM
12	As a user, i want real-time updates of my emission data	Unassigned	New	Carbon Emission Management...			4/27/2026 4:33:28 PM
13	Carbon Footprint Analysis System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:33:16 PM
14	Carbon Footprints calculator	Unassigned	New	Carbon Emission Management...			4/27/2026 4:32:05 PM
15	Data Visualization(charts/graphs)	Unassigned	New	Carbon Emission Management...			4/27/2026 4:31:51 PM
16	Comparative Analysis	Unassigned	New	Carbon Emission Management...			4/27/2026 4:31:37 PM
17	As a user, i want to view my total carbon footprint	Unassigned	New	Carbon Emission Management...			4/27/2026 4:31:22 PM
18	As a user, i want to see my emissions in graphical form	Unassigned	New	Carbon Emission Management...			4/27/2026 4:31:09 PM
19	As a user, i want to compare my emissions over time	Unassigned	New	Carbon Emission Management...			4/27/2026 4:30:57 PM
20	Eco-Block Recommendation System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:30:43 PM
21	Eco-Friendly Suggestions	Unassigned	New	Carbon Emission Management...			4/27/2026 4:30:19 PM
22	Materials Recommendation	Unassigned	New	Carbon Emission Management...			4/27/2026 4:30:05 PM
23	Reduction Strategies	Unassigned	New	Carbon Emission Management...			4/27/2026 4:29:47 PM
24	As a user, i want suggestion to reduce my carbon emissions	Unassigned	New	Carbon Emission Management...			4/27/2026 4:29:26 PM
25	As a user, i want recommendation based on my usage	Unassigned	New	Carbon Emission Management...			4/27/2026 4:29:06 PM
26	Analytics and Reporting System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:28:40 PM
27	Report Generation	Unassigned	New	Carbon Emission Management...			4/27/2026 4:27:33 PM

Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

To learn about how to create epics, user story, features, backlogs for Carbon Emission Management System project.

Create Epic, Features, User Stories, Task

The screenshot shows the Azure DevOps interface for a project named 'Carbon Emission Management'. The 'Work Items' view is active, displaying a list of 27 items. The items are organized into a table with columns for ID, Title, Assigned To, State, Area Path, Tags, Comments, and Activity Date. The items are categorized into various areas such as User Management, Carbon Emission Tracking, and Analytics. The first item, 'User Management System', is highlighted.

ID	Title	Assigned To	State	Area Path	Tags	Comments	Activity Date
1	User Management System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:36:01 PM
2	User Registration	Unassigned	New	Carbon Emission Management...			4/27/2026 4:35:53 PM
3	User login and authentication	Unassigned	New	Carbon Emission Management...			4/27/2026 4:35:39 PM
4	Profile Management	Unassigned	New	Carbon Emission Management...			4/27/2026 4:35:26 PM
5	As a user, i want to login securely using my credentials	Unassigned	New	Carbon Emission Management...			5/2/2026 3:27:21 AM
6	As a user, i want to update my profile information	Unassigned	New	Carbon Emission Management...			4/27/2026 4:34:55 PM
7	Carbon Emission Tracking System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:34:37 PM
8	Activity Data Input	Unassigned	New	Carbon Emission Management...			4/27/2026 4:34:27 PM
9	Emission Tracking Module	Unassigned	New	Carbon Emission Management...			4/27/2026 4:34:14 PM
10	Real-Time Data Update	Unassigned	New	Carbon Emission Management...			4/27/2026 4:33:52 PM
11	As a user, i want to enter my daily activities to track emissions	Unassigned	New	Carbon Emission Management...			4/27/2026 4:33:39 PM
12	As a user, i want real-time updates of my emission data	Unassigned	New	Carbon Emission Management...			4/27/2026 4:33:28 PM
13	Carbon Footprint Analysis System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:33:16 PM
14	Carbon Footprints calculator	Unassigned	New	Carbon Emission Management...			4/27/2026 4:32:05 PM
15	Data Visualization(charts/graphs)	Unassigned	New	Carbon Emission Management...			4/27/2026 4:31:51 PM
16	Comparative Analysis	Unassigned	New	Carbon Emission Management...			4/27/2026 4:31:37 PM
17	As a user, i want to view my total carbon footprint	Unassigned	New	Carbon Emission Management...			4/27/2026 4:31:22 PM
18	As a user, i want to see my emissions in graphical form	Unassigned	New	Carbon Emission Management...			4/27/2026 4:31:09 PM
19	As a user, i want to compare my emissions over time	Unassigned	New	Carbon Emission Management...			4/27/2026 4:30:57 PM
20	Eco-Block Recommendation System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:30:43 PM
21	Eco-Friendly Suggestions	Unassigned	New	Carbon Emission Management...			4/27/2026 4:30:19 PM
22	Materials Recommendation	Unassigned	New	Carbon Emission Management...			4/27/2026 4:30:05 PM
23	Reduction Strategies	Unassigned	New	Carbon Emission Management...			4/27/2026 4:29:47 PM
24	As a user, i want suggestion to reduce my carbon emissions	Unassigned	New	Carbon Emission Management...			4/27/2026 4:29:26 PM
25	As a user, i want recommendation based on my usage	Unassigned	New	Carbon Emission Management...			4/27/2026 4:29:06 PM
26	Analytics and Reporting System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:28:40 PM
27	Report Generation	Unassigned	New	Carbon Emission Management...			4/27/2026 4:27:11 PM

1. Fill in Epics

The screenshot shows the Azure DevOps interface for the same project, but with the 'Work Items' view filtered to show only items categorized as 'epic'. The list contains 5 items, all with the state 'New' and assigned to 'Unassigned'. The items are: 'User Management System', 'Carbon Emission Tracking System', 'Carbon Footprint Analysis System', 'Eco-Block Recommendation System', and 'Analytics and Reporting System'.

ID	Title	Assigned To	State	Area Path	Tags	Comments	Activity Date
1	User Management System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:36:01 PM
7	Carbon Emission Tracking System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:34:37 PM
13	Carbon Footprint Analysis System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:33:16 PM
20	Eco-Block Recommendation System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:30:43 PM
26	Analytics and Reporting System	Unassigned	New	Carbon Emission Management...			4/27/2026 4:28:40 PM

2.Fill in User Story Details

Azure DevOps krishnakumarm2024cse / Carbon Emission Managem... / Boards / Work items

Search

Carbon Emission Mana... +

Overview

Boards

Work Items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Pipelines

Artifacts

Project settings

Work items

Recently updated + New Work Item Open filtered view in Queries Column Options Import Work Items Recycle Bin

user story

ID	Title	Assigned To	State	Area Path	Tags	Comments	Activity Date
5	As a user, i want to login securely using my credentials	Unassigned	New	Carbon Emission Management...			5/2/2026 3:27:21 AM
6	As a user, i want to update my profile information	Unassigned	New	Carbon Emission Management...			4/27/2026 4:34:55 PM
11	As a user, i want to enter my daily activities to track emissions	Unassigned	New	Carbon Emission Management...			4/27/2026 4:33:39 PM
12	As a user, i want real-time updates of my emission data	Unassigned	New	Carbon Emission Management...			4/27/2026 4:33:28 PM
17	As a user, i want to view my total carbon footprint	Unassigned	New	Carbon Emission Management...			4/27/2026 4:31:22 PM
18	As a user, i want to see my emissions in graphical form	Unassigned	New	Carbon Emission Management...			4/27/2026 4:31:09 PM
19	As a user, i want to compare my emissions over time	Unassigned	New	Carbon Emission Management...			4/27/2026 4:30:57 PM
24	As a user, i want suggestion to reduce my carbon emissions	Unassigned	New	Carbon Emission Management...			4/27/2026 4:29:26 PM
25	As a user, i want recommendation based on my usage	Unassigned	New	Carbon Emission Management...			4/27/2026 4:29:06 PM
30	As a user, i want to view detailed analytics	Unassigned	New	Carbon Emission Management...			4/27/2026 4:26:17 PM
31	As a user, i want to download reports for future reference	Unassigned	New	Carbon Emission Management...			4/27/2026 4:25:55 PM
32	As a user, i want to login securely using my credentials	Naren S	Active	Carbon Emission Management...			4/29/2026 5:12:38 PM
33	As a user, i want to update my profile information	Naren S	Active	Carbon Emission Management...			5/2/2026 3:27:04 AM
34	As a user, i want to enter my daily activities to track emissions	Naren S	Active	Carbon Emission Management...			4/29/2026 5:12:47 PM
35	As a user, i want real-time updates of my emission data	Unassigned	New	Carbon Emission Management...			4/29/2026 5:10:54 PM
36	As a user, i want to view my total carbon footprint	Unassigned	New	Carbon Emission Management...			4/29/2026 5:11:05 PM
37	As a user, i want to see my emissions in graphical form	Unassigned	New	Carbon Emission Management...			4/29/2026 5:11:10 PM
38	As a user, i want to compare my emissions over time	Unassigned	New	Carbon Emission Management...			4/29/2026 5:11:22 PM

Result:

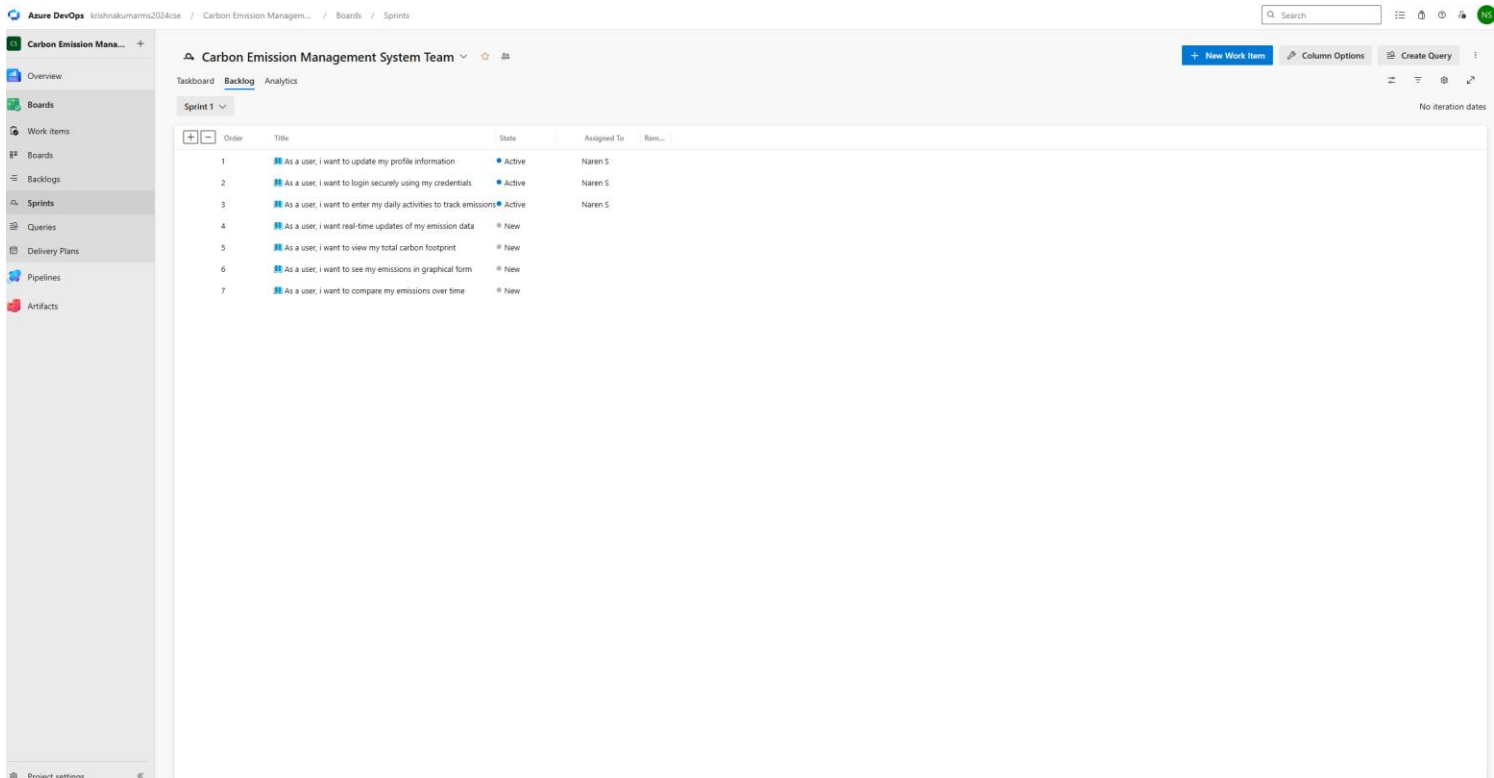
Thus, the creation of epics, features, user story and task has been created successfully.

SPRINT PLANNING

Aim:

To assign user story to specific sprint for Carbon Emission Management System Project.

Sprint Planning



Sprint Planning – Description

Sprint Planning is a key meeting in Agile development where the team decides what work will be completed in the upcoming sprint. During this session, the product owner presents the prioritized tasks (user stories), and the team discusses, estimates, and selects the items they can realistically finish within the sprint duration.

It helps the team set clear goals, plan their work, and ensure everyone understands their responsibilities. Overall, Sprint Planning ensures that the sprint starts with a clear direction and achievable object.

Result:

The Sprints are created for Carbon Emission Management System Project.

POKER ESTIMATION

Aim:

Create Poker Estimation for the user stories – Carbon Emission Management System Project.

Poker Estimation

The screenshot displays the Azure DevOps interface for a project named "Carbon Emission Management System". The main view is a user story titled "As a user, I want to login securely using my credentials". The story is assigned 5 story points. The interface includes a sidebar with navigation options like Overview, Boards, Work items, Backlogs, Sprints, Queries, Delivery Plan, Pipelines, and Artifacts. The main content area is divided into several sections: Description, Planning, Deployment, Development, and Related Work. The Description section contains a detailed description of the login functionality and a list of acceptance criteria. The Planning section shows the story points (5), priority (2), and risk. The Deployment section provides instructions on how to track releases. The Development section includes a link to the Azure Repos commit pull request or branch. The Related Work section shows a link to the User Management System. A comment box is visible at the bottom of the story, with a placeholder text "Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request."

USER STORY 5
5 As a user, I want to login securely using my credentials

0 Comments Add Tag

Save and Close Follow Details

Updated by Naren S. May 2

Description

The system should provide a secure login functionality that allows registered users to access their accounts using valid credentials such as email and password.

The login process must ensure authentication, protect user data, and prevent unauthorized access. It should include validation checks, error handling, and basic security measures like password encryption.

Acceptance Criteria

- The system should allow users to enter email/username and password
- The system should validate credentials against stored user data
- The user should be successfully logged in with valid credentials
- The system should display an error message for invalid credentials
- Password input should be masked (hidden)
- The system should prevent login without required fields
- The system should ensure secure handling of user data (encryption/authentication)
- The user should be redirected to the dashboard after successful login

Planning

Story Points: 5
Priority: 2
Risk:

Classification

Value area: Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Parent

1 User Management System
Updated Apr 27 @ New

Project settings

Result:

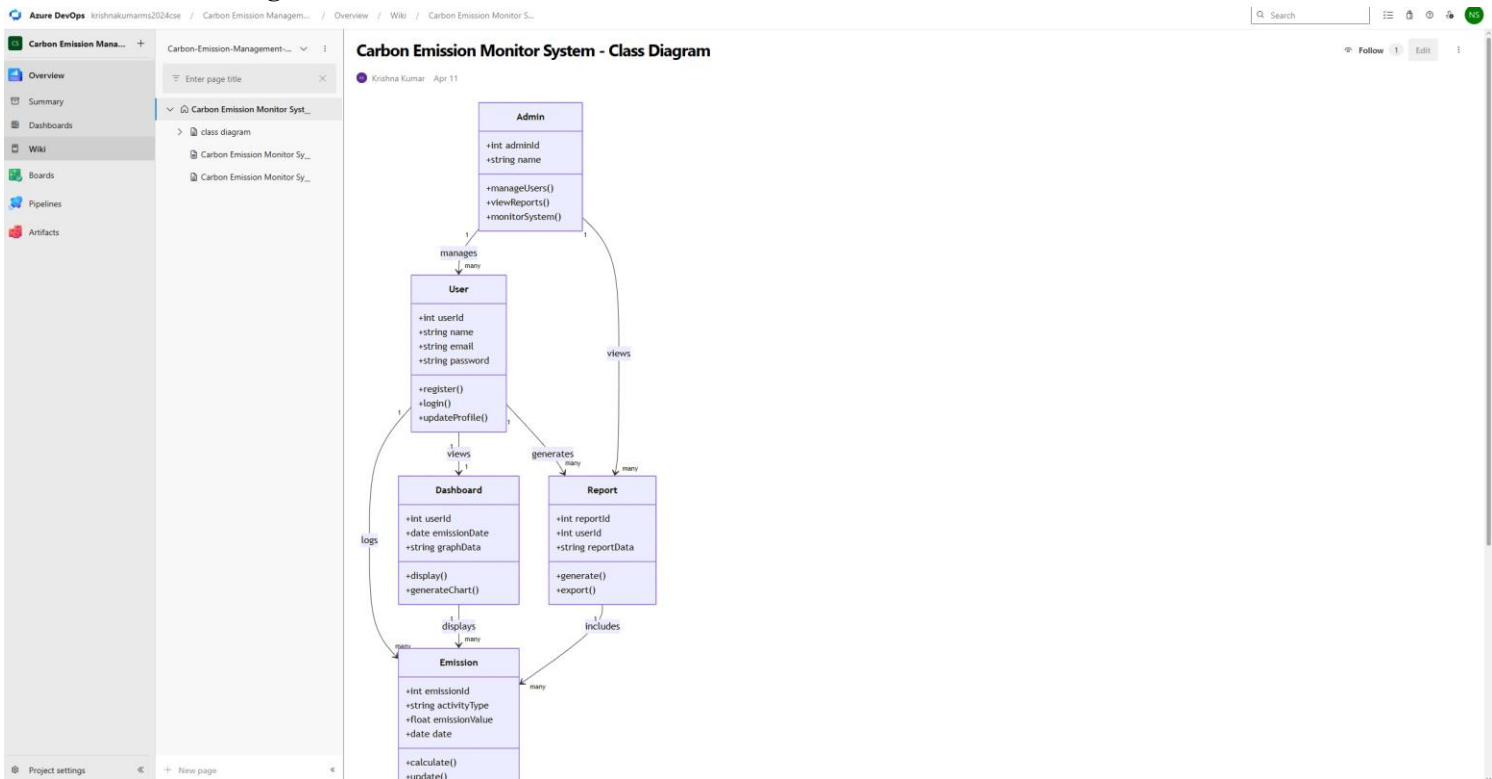
The Estimation/Story Points is created for the project using Poker Estimation.

DESIGNING CLASS, SEQUENCE AND USE CASE DIAGRAMS FOR PROJECT ARCHITECTURE

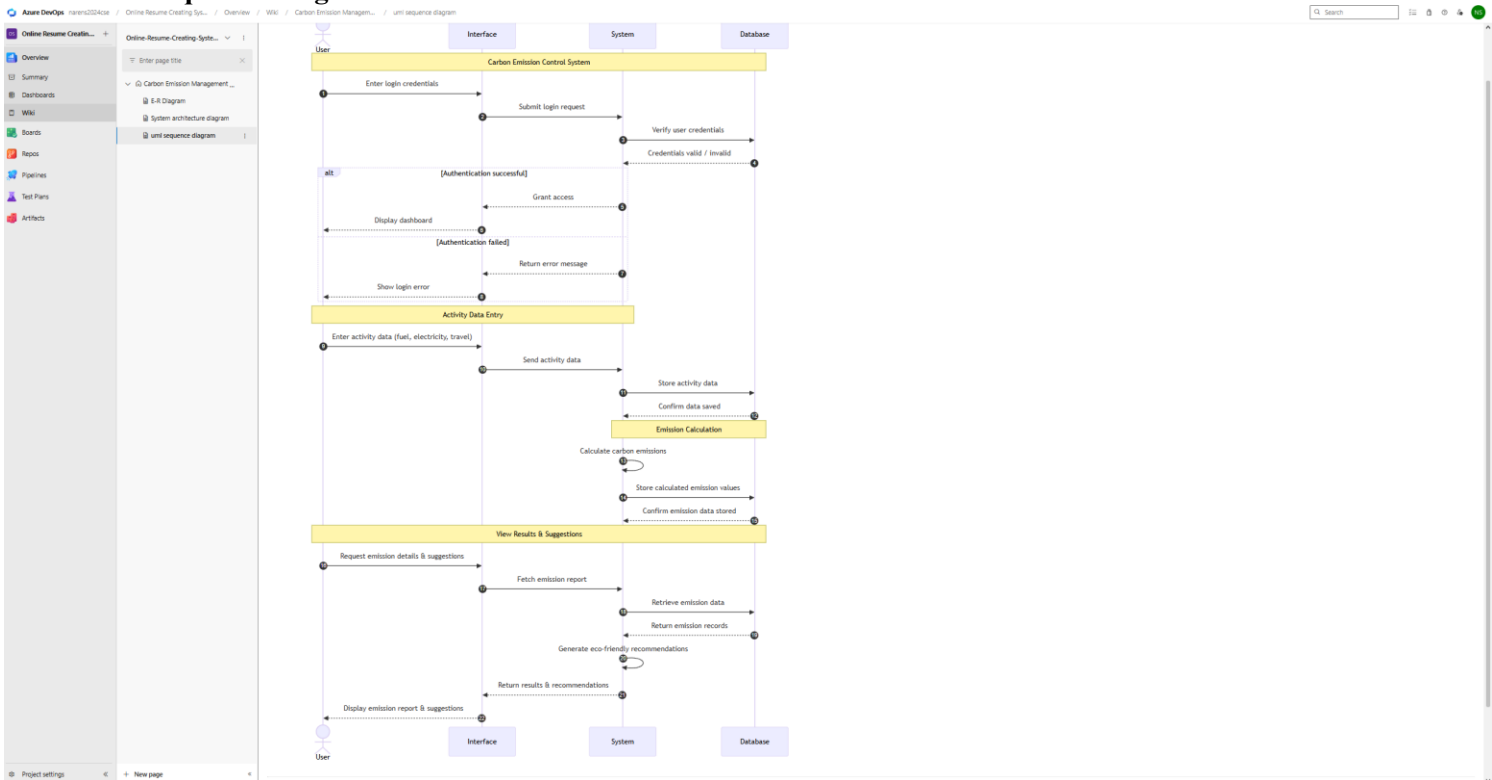
Aim:

To Design a Class Diagram, Sequence Diagram and Use Case Diagram for Online Job Application System Project.

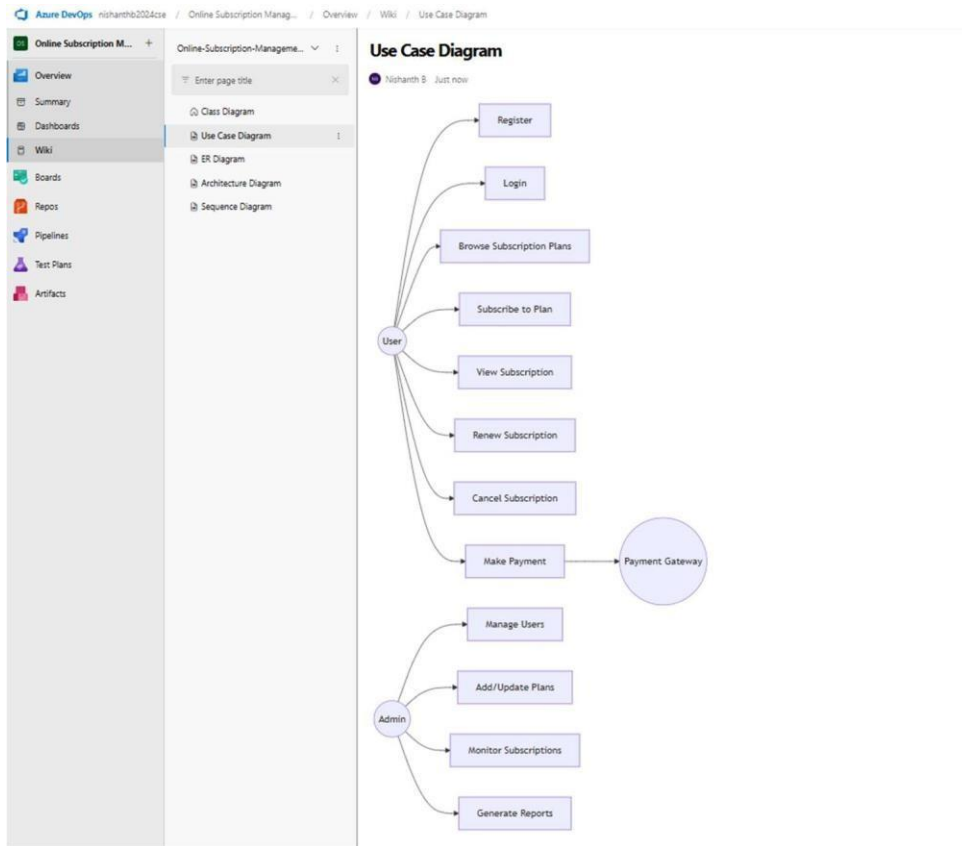
6A. Class Diagram



6B. Sequence Diagram



6C. Use Case Diagram



Result:

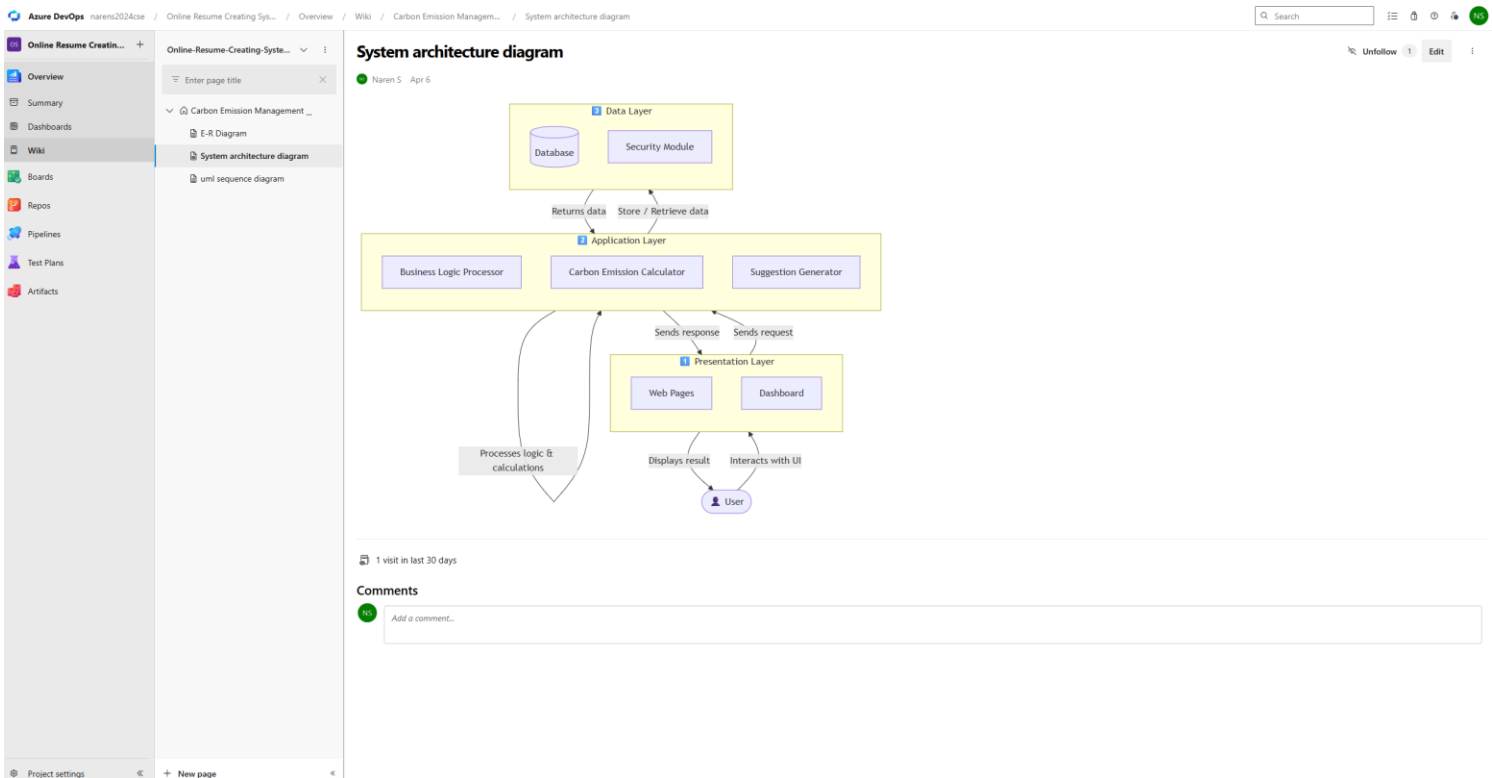
The Class Diagram and Sequence Diagram is designed Successfully for Carbon Emission Management System.

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

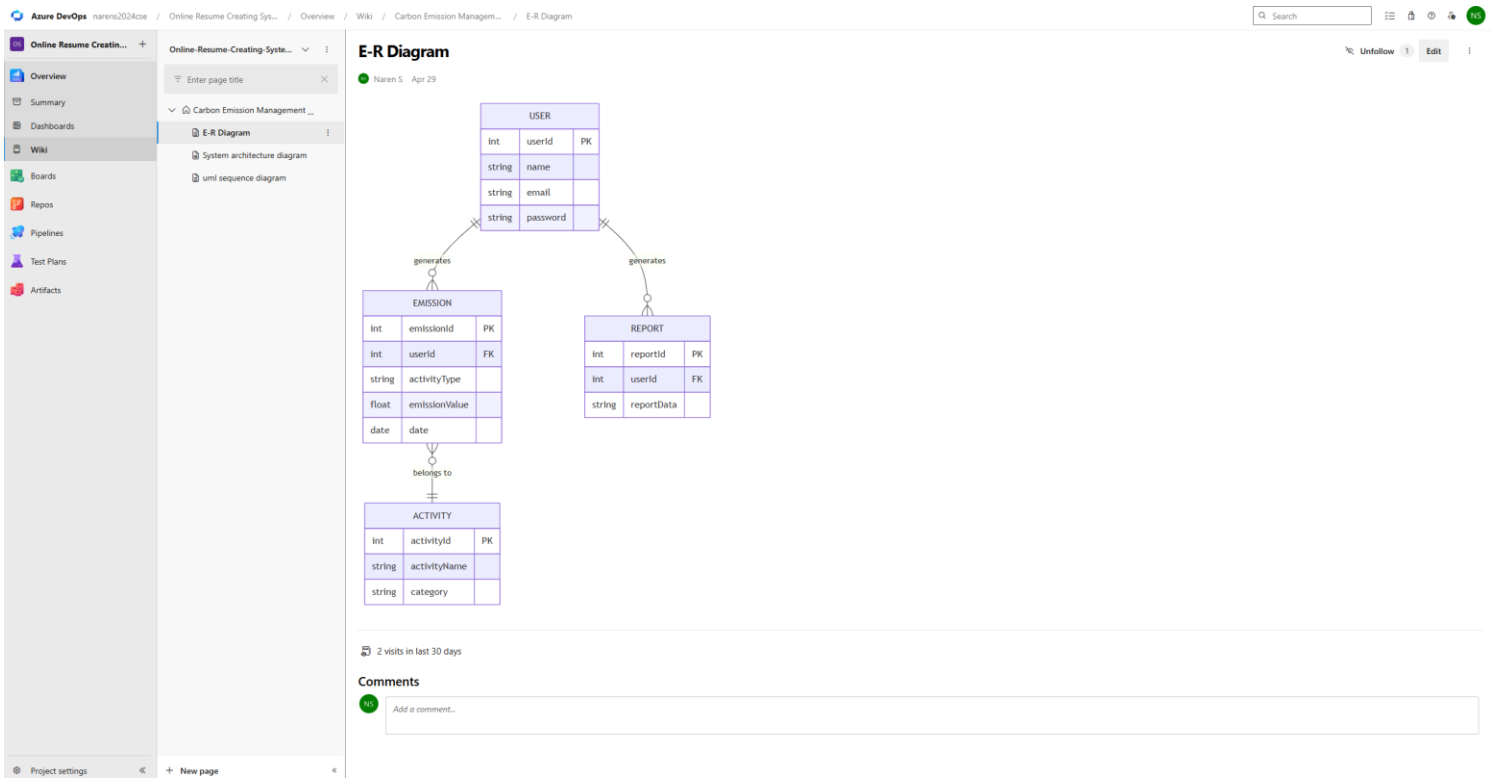
Aim:

To Design an Architectural Diagram and ER Diagram for Carbon Emission Management System Project.

7A. Architectural Diagram



7B.ER Diagram.



Result:

The Architecture Diagram and ER Diagram is designed Successfully for Carbon Emission Management System Project.

TESTING – TEST PLANS AND TEST CASES

Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case

1.New test plan

The screenshot shows the Azure DevOps interface for a test plan. The left sidebar contains navigation links: Overview, Boards, Repos, Pipelines, Test Plans, Test plans, Progress report, Parameters, Configurations, Runs, Exploratory sessions, and Artifacts. The main content area is titled 'Carbon Emission Control System - Test Plan (ID: 4)' and includes tabs for Define, Execute, and Chart. Below the title, there is a section for 'Test Suites' with a filter 'Filter suites by name'. The list of test suites includes 'Carbon Emission Control System - Test Suite (1)', 'Login Test Suite (2)', 'Emission Test Suite (2)', and 'Dashboard Test Suite (1)'. In the center of the page, there is an illustration of a person writing on a whiteboard, with the text 'Add a test case' and a button labeled 'New Test Case'.

2. Test suite

The screenshot shows the Azure DevOps interface for a test suite. The left sidebar contains navigation links: Overview, Boards, Repos, Pipelines, Test Plans, Test plans, Progress report, Parameters, Configurations, Runs, Exploratory sessions, and Artifacts. The main content area is titled 'Login Test Suite (ID: 5)' and includes tabs for Define, Execute, and Chart. Below the title, there is a section for 'Test Cases (2 items)' with a table of test cases. The table has columns: Title, Order, Test Case Id, Assigned To, and State. The test cases are 'User Login with valid credentials' (Order 1, Test Case Id 8, Assigned To Naren S, State Design) and 'Invalid Login with wrong password' (Order 2, Test Case Id 9, Assigned To Naren S, State Design). A 'New Test Case' button is located at the top right of the table.

Title	Order	Test Case Id	Assigned To	State
User Login with valid credentials	1	8	Naren S	Design
Invalid Login with wrong password	2	9	Naren S	Design

3. Test case

The screenshot displays the Azure DevOps Test Case interface. The left sidebar shows the navigation menu with options like Overview, Boards, Repos, Pipelines, Test Plans, and Artifacts. The main area is titled 'TEST CASE 8' and 'User Login with valid credentials'. It shows the test case details, including the author 'Naren S', comments, and tags. The 'Steps' section lists three actions: 'Enter valid username and password', 'Click the Login button', and 'Observe the page after login'. The 'Expected result' column shows 'Login successful, dashboard loads', 'Login button responds', and 'Dashboard loads successfully'. The right sidebar contains sections for 'Recent test results', 'Deployment', 'Development', 'Related Work', and 'Status'.

TEST CASE 8
User Login with valid credentials

Author: Naren S
Comments: 0
Add Tag

Save and Close | Follow | Updated by Naren S: Apr 29

Steps | Summary | Associated Automation

Steps

Steps	Action	Expected result	Attachments
1.	Enter valid username and password	Login successful, dashboard loads	
2.	Click the Login button	Login button responds	
3.	Observe the page after login	Dashboard loads successfully	

Click or type here to add a step

Recent test results
Run manual tests with web runner. [Learn how to get started](#)

Deployment
To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development
Add link
Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work
Add link
[Add an existing work item as a parent](#)

Status
Priority: 1
Automation status: Not Automated

Parameter values

Test report summary

The screenshot displays the Azure DevOps Test Plans interface. The left sidebar contains navigation links: Overview, Boards, Repos, Pipelines, Test Plans, Test plans, Progress reports, Parameters, Configuration, Runs, Exploratory tests, and Artifacts. The main area shows a test case titled "8 User Login with valid credentials" created by Naren S. The test case is associated with the "Online Resume Creating System" area and the "Online Resume Creating System(Sprint 1)" iteration. The test steps are as follows:

Steps	Action	Expected result	Attachments
1.	Enter valid username and password	Login successful, dashboard loads	
2.	Click the Login button	Login button responds	
3.	Observe the page after login	Dashboard loads successfully	

Below the steps, there is a link to "Click or type here to add a step". The right sidebar shows the "Recent test results" section with a link to "Run manual tests with web runner". Below this, the "Deployment" section provides instructions on how to track releases and turn on deployment status reporting. The "Development" section includes a link to "Add link" and a note about linking an Azure Repos commit, pull request, or branch. The "Related Work" section has a link to "Add link" and a note about adding an existing work item as a parent. The "Status" section shows the priority as 1 and the automation status as "Not Automated".

Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

	LOAD TESTING AND PIPELINES
--	-------------------------------------

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
2. Create the Resource
 - o Go to *Create a resource* → Search for “Azure Load Testing”.
 - o Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
 - o *Subscription*: Choose your Azure subscription.
 - o *Resource Group*: Create new or select an existing one.
 - o *Name*: Provide a unique name (no special characters).
 - o *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

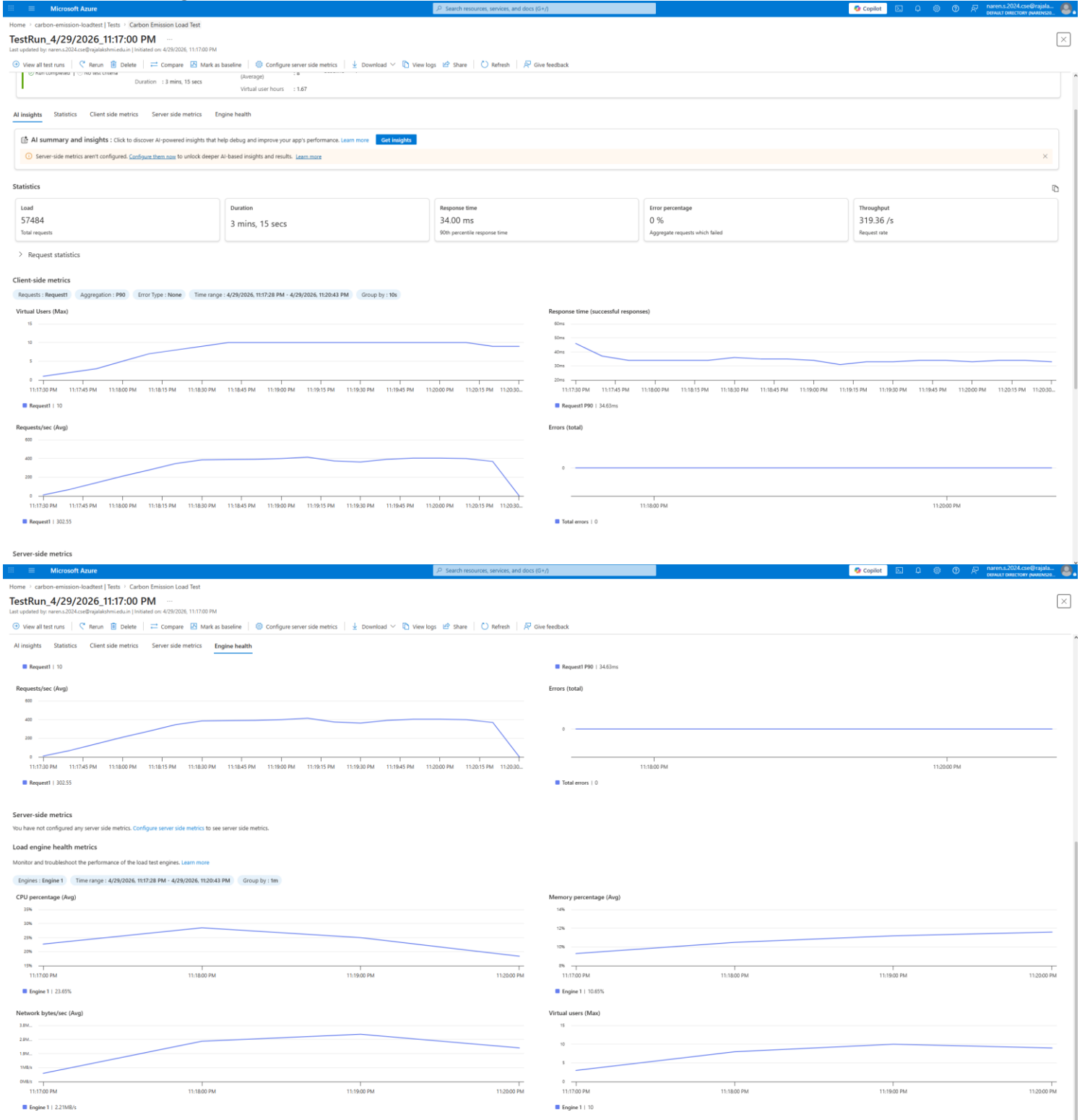
Steps to Create and Run a Load Test:

Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab
 - o *Test Name*: Provide a unique name.
 - o *Description*: (Optional) Add test purpose.
 - o *Run After Creation*: Keep checked.
3. Load Settings
 - o *Test URL*: Enter the target endpoint (e.g., <https://yourapi.com/products>).

4. Click Review + Create → Create to start the test.

Load Testing



Pipelines

Description:

This experiment demonstrates how to connect a GitHub-hosted project with Azure DevOps. The pipeline automatically installs dependencies, builds the application, and runs basic scripts to verify functionality. This ensures that every commit triggers automated checks, improving reliability and enabling smooth deployment.

Steps:

1. **Connect GitHub to Azure DevOps:**
 - Create a new project in Azure DevOps
 - Create a pipeline and select GitHub as the source
 - Authorize access to your GitHub repository
2. **Create azure-pipelines.yml in repository root:**
 - Add a new file named azure-pipelines.yml
 - Paste the configuration below
3. **Run and Monitor Pipeline:**
 - Commit code to the main branch
 - Pipeline will trigger automatically
 - Check logs in Azure DevOps for execution status

Pipeline Tasks Include:

- Setting up Node.js environment using NodeTool
- Installing frontend and backend dependencies
- Building frontend application
- Running backend script to verify execution
- Automatically triggering pipeline on every commit

Pipeline

The screenshot displays the Azure DevOps web interface for a pipeline run. The left sidebar contains navigation links: Overview, Boards, Repos, Pipelines (selected), Environments, Library, Test Plans, and Artifacts. The main content area shows a successful build #20260427.1 for the pipeline 'Set up CI with Azure Pipelines'. A message indicates the run is retained as a recent run. The 'Summary' section provides details about the build, including the repository 'Nimalan240701362/Online Subscription Management System', the time started and elapsed (Yesterday at 6:42 PM, 9s), and related work items and artifacts. The 'Stages' section shows a single stage 'Stage' with 1 job completed in 7s, and a 'Job' completed in 3s. A 'Rerun Stage' button is visible.

Azure DevOps

Search

Online Subscription Manag... +

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Library

Test Plans

Artifacts

Project settings

#20260427.1 • Set up CI with Azure Pipelines
Nimalan240701362/Online-Subscription-Management-System

This run is being retained as one of recent runs by pipeline. [View retention leases](#)

Summary

Individual CI by Nimalan240701362

Repository and version
Nimalan240701362/Online Subscription Management System
main 32000fb

Time started and elapsed
Yesterday at 6:42 PM
9s

Related
0 work items
0 artifacts

Tests and coverage
[Get started](#)

[Parameters](#)

Stages Jobs

Stage
1 job completed 7s

Job
3s

[Rerun Stage](#)

Result:

Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend for Online Job Application System project.

GitHub Project Structure

Files

main

Go to file

./kino/specs/enhanced-street-mode-simulation

designs.md

requirements.md

tasks.md

./src

backend

mobile-app

web-client

gitattributes

CREATIVE_KIRO_USAGE.md

Software-Construction-Project_Carbon-Emission-Management-System

Initial commit

144518c - 35 minutes ago

History

Preview Code Blame 488 lines (283 loc) - 38.8 KB

Raw

Design Document: Enhanced Street Mode Simulation

Overview

This design enhances the EcoBlocks street mode visualization system to provide realistic, immersive environmental intervention simulations. The current implementation suffers from poor camera positioning, barely visible particle effects, and imprecise heatmaps that cover large areas rather than individual buildings. This design addresses these issues through advanced CesiumJS features, optimized particle systems, and building-level precision visualization.

The enhanced system will leverage Cesium OSM Buildings for detailed 3D building models, implement sophisticated particle systems for realistic air quality visualization, and provide precise building-level pollution mapping with interactive intervention placement.

Architecture

System Components

Diagram showing a hierarchical tree structure of system components.

Data Flow

Diagram showing the flow of data between components: User, Street Mode Controller, Camera System, Building Visualizer, Particle System, Heatmap System, and Intervention System.

Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.